$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\frac{\partial f}{\partial x} \quad \frac{\partial^2 f}{\partial x \partial y} \quad \frac{\partial^2 f}{\partial x^2} \quad \frac{\partial^3 f}{\partial x^2 \partial y} \quad \frac{\partial^3 f}{\partial x \partial y \partial x} \quad \frac{\partial}{\partial t} \quad \frac{\partial^2 u}{\partial x_1^2}$$

$$H_f = \begin{bmatrix} \dfrac{\partial^2 f(v)}{\partial x_1^2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_2^2} \end{bmatrix}$$

$$H_f = \begin{bmatrix} \dfrac{\partial^2 f(v)}{\partial x_1^2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_3} \\ \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_2^2} & \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_3} \\ \dfrac{\partial^2 f(v)}{\partial x_3 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_3 \partial x_2} & \dfrac{\partial^2 f(v)}{\partial x_3^2} \end{bmatrix}$$

$$H_g = \begin{bmatrix} \dfrac{\partial^2 g(v)}{\partial y_1^2} & \dfrac{\partial^2 g(v)}{\partial y_1 \partial y_2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_1 \partial y_n} \\ \dfrac{\partial^2 g(v)}{\partial y_2 \partial y_1} & \dfrac{\partial^2 g(v)}{\partial y_2^2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_2 \partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \dfrac{\partial^2 g(v)}{\partial y_n \partial y_1} & \dfrac{\partial^2 g(v)}{\partial y_n \partial y_2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_n^2} \end{bmatrix}$$

$$H_f(\mathbf{a}) = \begin{bmatrix} \dfrac{\partial^2 f(\mathbf{a})}{\partial x_1^2} & \dfrac{\partial^2 f(\mathbf{a})}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f(\mathbf{a})}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(\mathbf{a})}{\partial x_2^2} \end{bmatrix} \qquad H_f = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$